

Component	Role in the System	Why We Used It
ESP32 DevKit	The Brain (MCU)	Dual-core power to handle real-time sensor polling on one core and WiFi/NTP/Firebase tasks on the other. It's the standard for IoT.
PMS5003	Laser Dust Sensor	Unlike cheap IR sensors, this uses laser scattering to count individual particles PM2.5 and PM10, providing stable, high-precision data.
AHT20/25	Climate Monitor	Offers a digital I2C interface with better long-term stability and faster response times than the older DHT11/22 series.
MQ-2	Smoke/Gas Guard	Essential for detecting combustible gases and smoke. It adds a "safety" layer to the environmental monitoring.
MQ-7	Carbon Monoxide (CO)	Specifically tuned for CO detection. Since CO is odorless and deadly, this is a critical industrial-grade metric for indoor safety.
16x2 LCD (I2C)	Local Interface	Provides an instant, "tactical" readout at the node site. Using the I2C module saves 4-6 GPIO pins, keeping your wiring clean for the freeform build.

